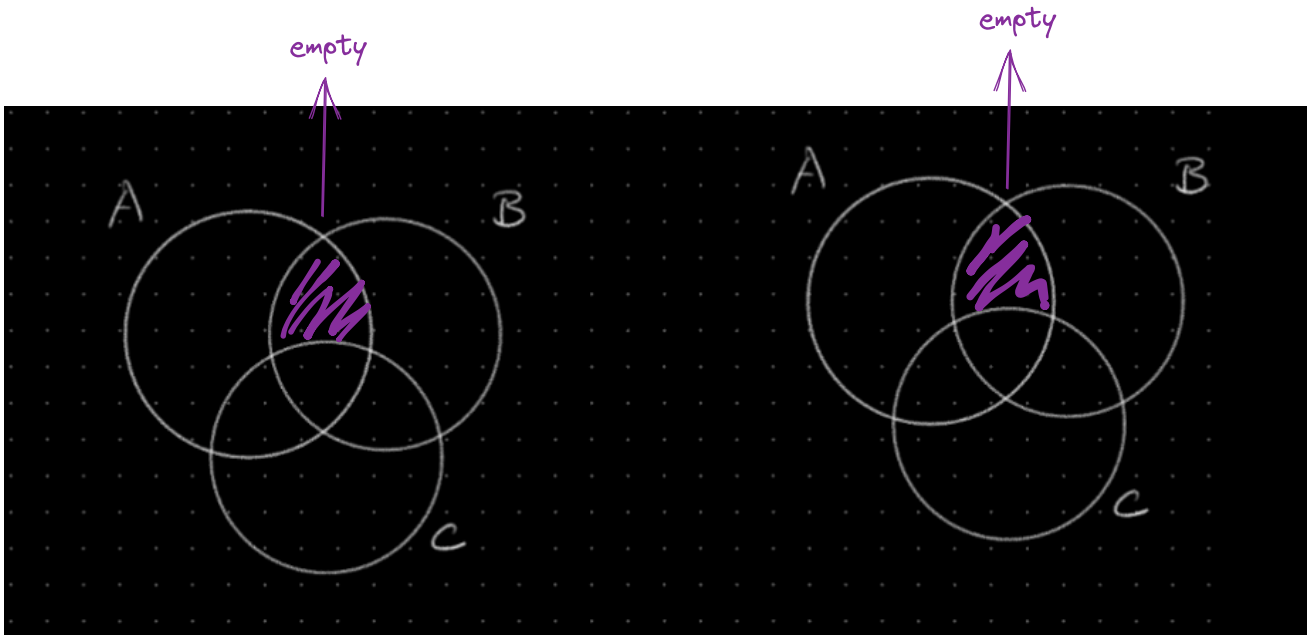
BCS1130-6 Set theory

Lecture notes

Proof example

Proof example

For all sets A, B, C, $(B \cap (A^c \cup C)^c = \phi) \iff (A \subseteq B^c \cup C)$



We try to prove that the statement is true (which we identified using Venn Diagrams)

First implication

$B \cap A \cap C^c = \phi \implies A \subseteq B^c \cup C$

- Let $x \in A$
  - Case 1: $x \in C$
    - $\implies x \in C \vee x \notin B \implies x \in B^c \cup C$
  - Case 2: $x \notin C$
    - Assume $x \in B \implies x \in A, x \in B \wedge x \notin C$
    - $x \in (B \cap A \cap C^c) \rightarrow$ This is a contradiction as that set is empty
    - Therefore $x \notin B \rightarrow x \in B^x \vee x \in C \implies x \in B^c \cup C$

Second implication

$A \subseteq B^c \cup C \implies B \cap A \cap C^c = \phi \rightarrow$ We can either take the contrapositive, or take a **contradiction**

Proof by contradiction

- We need to assume the set $B \cap A \cap C^c$ is not empty → there is an element in that set: $x \in B \cap A \cap C^c$
  - then $x \in B, x \in A$ and $x \notin C$
  - so $(x \in B^c \vee x \in C) \wedge (x \in B \vee x \notin C)$ → Contradiction (the two sets are opposite)
  - so there is no $x \in B \cap A \cap C^c$ → the set $B \cap A \cap C^c$ is **empty**

Proof example (shorter)

Proof example

For all sets A, B, C, $(B \cap (A^c \cup C)^c = \phi) \iff (A \subseteq B^c \cup C)$

We prove: $\neg(A \subseteq B^c \cup C) \iff \neg(B \cap (A^c \cup C)^c = \phi)$

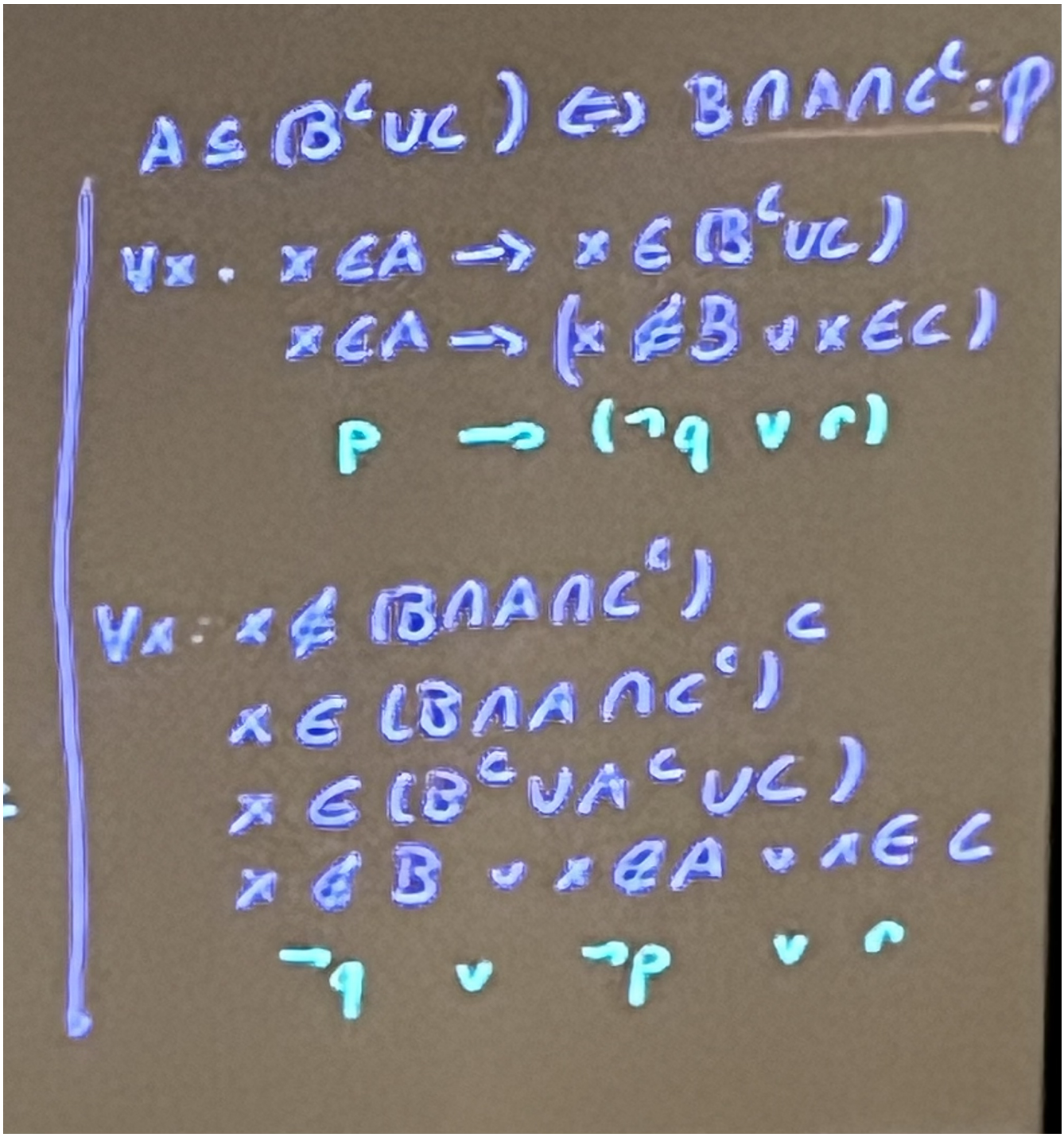
Subset: $\forall x \in A : x \in B$
Not subset: $\exists x \in A : x \notin B$

- $\neg(A \subseteq (B^c \cup C))$
  - $\iff \exists x \in A : x \notin (B^c \cup C)$
  - $\iff \exists x \in A : x \in (B^c \cup C)^c$ (if an element is not in something, it is an element of the complement)
  - $\iff \exists x \in A : x \in (B \cap C^c)$
  - $\iff \exists x : x \in A \wedge x \in B \wedge x \in C^c$
  - $\iff \exists x : x \in (A \cap B \cap C^c)$ → X is in the intersection of the three sets
  - $\iff (A \cap B \cap C^c) \neq \phi$ → The set is not empty

Proof example (truth table)

Proof example

For all sets A, B, C, $(B \cap (A^c \cup C)^c = \phi) \iff (A \subseteq B^c \cup C)$



Power set

Power sets

A power set of a set A is the set of all subsets of A

Example of power sets

Example of a power set:
$A = \{1, 2, 3\}$
$P(A) = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{\}, \{1, 2, 3\}\}$
$|P(A)| = 8$ (set size is 8)

Example of a power set:
$B = \{i, \phi\}$
$P(B) = \{\phi, \{1, \phi\}, \{1\}, \{\phi\}\}$

Cardinality of a power set

Size of a power set

$$|P(A)| = 2^{|A|}$$

Product sets

Product sets

For two sets A and B, the product set $A \times B$ is defined as
$\{(a, b)$ with $a \in A \wedge b \in B\}$

- (a, b) is an ordered pair

Example of product sets

$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} \rightarrow$ Cartesian plane

Cardinality of product sets

Cardinality of product sets

$$|A \times B| = |A| \times |B|$$

Properties

- $A \times B = B \times A$
  - when $A = B$
  - when one of the sets is empty $\rightarrow$ The product is also empty

Set partitions

Definition

Set partitions

The sets $A_1, \ldots A_n$ form a partition of the set A $\iff$

1. None of the sets A_k is empty: $A_k \neq \forall k$
2. $A_1 \cup A_2 \ldots \cup A_n = \cup_{k=1}^{n} A_k = A$
3. $A_m \cap A_k = \phi$ if $m \neq k$

How many partitions are there

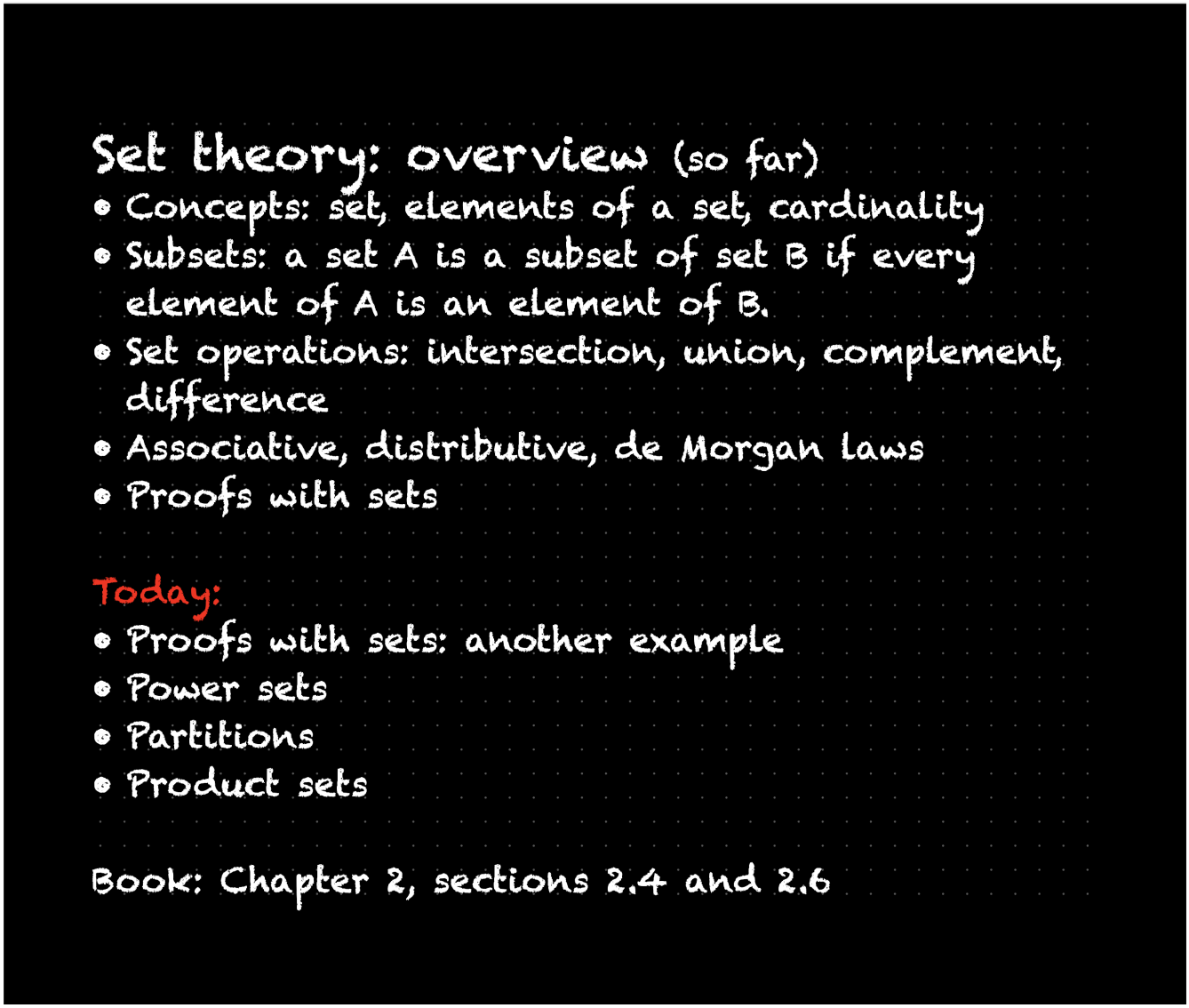
Partitions

$$B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_k$$

Lecture slides

- [ ] Replace with annotated notes

Lecture slides



Partitions

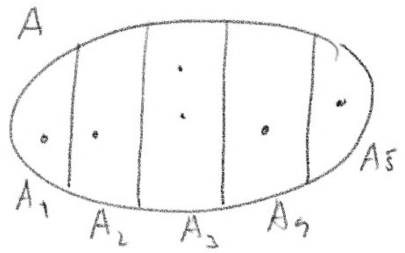
- PARTITION

The sets $A_1, \dots, A_n$ form a partition of the set A iff

1) none of the sets A_k is empty $A_k \neq \emptyset \quad \forall k$

2) $A_1 \cup A_2 \dots \cup A_n = \bigcup_{k=1}^{n} A_k = A$

3) $A_m \cap A_k = \emptyset$ if $m \neq k$



$A = \{1, 2, 3\}$

$A_1 = \{1\}$, $A_2 = \{2, 3\}$ is a partition